Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 19
Matrix Inverses.

Review

1. QR Factorization (Sec. 10.4)

Matrix Inverses!

1. Check that C^T is a right-inverse of A^T (so $A^T C^T = I$) for:

$$A = \begin{bmatrix} -3 & -4 \\ 4 & 6 \\ 1 & 1 \end{bmatrix}$$

$$C = \frac{1}{2}\begin{bmatrix} 0 & -1 & 6 \\ 0 & 1 & -4 \end{bmatrix}$$

2. Let $\vec{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Write the equations that correspond to the system of linear equations

$$A^T \vec{x} = \vec{b}.$$

3. Use the fact that C^T is a right inverse of A^T to find a solution of this system. Check your solution.

4. If a matrix A^T has a right-inverse then its rows must be linearly independent. Check that the rows of A^T (last example) are linearly independent.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$B = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

5. Check that B is a left-inverse of A (so $BA = I$).

6. Check that B is a right-inverse of A (so $AB = I$).

7. Find A^{-1}.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

8. Now use the 2×2 inverse formula to find A^{-1}.